

- This exam contains 12 pages (including the front and back of this cover page) and 14 problems. Check to see if any pages are missing.
- This is the final exam and will count for 20% of your final grade. You have 120 minutes (3:30 pm - 5:30 pm) to complete this exam. You may use one index card of notes and a calculator on this exam, but no other outside sources may be consulted.

The following rules apply:

- Every problem on this exam, other than true-or-false problems, requires work of some form. An incorrect answer supported by substantially correct calculations and explanations will receive partial credit. **A correct answer that has no supporting work will not receive full credit.**
- Organize your work, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little to no credit.
- There are 100 points total in this exam, as well as 1 bonus questions worth a total of 5 possible points.

Page:	3	4	5	6	7	8	9	10	11	Total
Points:	10	10	10	10	8	8	14	20	10	100
Score:										

Bonus:	
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1. Mark each statement as ☐ True or ☐ False by completely shading in the box corresponding to your answer.

2 pts

- (a) The conic section represented by the equation $4x^2 + 16x - 16y^2 + 64y + 72 = 0$ is a parabola.

☐ True

☐ False

2 pts

- (b) The conic section represented by the equation $3x^2 + 5xy + 4y^2 + 1 = 0$ is an ellipse.

☐ True

☐ False

2 pts

- (c) An explicit formula for the arithmetic sequence (starting at a_1) given by $\{-2/3, -1, -4/3, \dots\}$ is $a_n = -(1/3)n - 1/3$.

☐ True

☐ False

2 pts

- (d) In summation notation, the geometric sum $1 + 4 + 16 + 64 + 256 + 1024 + 4096 + 16384$ can be expressed as $\sum_{n=0}^8 4^n$.

☐ True

☐ False

2 pts

- (e) The sequence $\{-1.4, 7, -35, 175, -875, \dots\}$ is geometric.

☐ True

☐ False

10 pts

2. Write the equation of the ellipse in standard form, and identify the end points of the major and minor axes as well as the foci.

$$25x^2 - 50x + 16y^2 - 96y - 231 = 0$$

Equation in Standard Form :

End Points of Major Axis :

End Points of Minor Axis :

Foci :

10 pts

3. Write the equation of the hyperbola in standard form, and identify the vertices and the foci.

$$16x^2 - 32x - 9y^2 - 36y - 164 = 0$$

Equation in Standard Form :

Vertices :

Foci :

10 pts

4. Rewrite the given equation in standard form, and then determine the vertex, focus, directrix, and axis of symmetry of the parabola.

$$2y^2 + 8y + 8x - 24 = 0$$

Equation in Standard Form : _____

Vertex : _____

Directrix : _____

Axis of Symmetry : _____

3 pts

5. Write the first six terms of the sequence.

$$a_1 = -2, \quad a_2 = 5, \quad a_n = a_{n-2}(4 - a_{n-1})$$

5 pts

6. Write the first five terms of the arithmetic sequence given the following two terms

$$a_{11} = -185, \quad a_{31} = -485$$

3 pts

7. Starting with a_1 , write the first five terms of the geometric sequence

$$a_n = -2(3)^{n-1}$$

5 pts

8. Find the 11th term in the arithmetic sequence below.

$$\{ -10, -13, -16, \dots \}$$

4 pts

9. Write an explicit formula for the geometric sequence

$$\left\{2, 1, \frac{1}{2}, \frac{1}{4}, \dots\right\}$$

10 pts

10. Find the value of the sum

$$1 + 2 + 3 + \dots + 999 + 1000.$$

10 pts 11. Find the value of the sum

$$-1 - \frac{1}{4} - \frac{1}{16} - \frac{1}{64} - \dots$$

10 pts 12. Find the sum of the finite geometric series

$$\sum_{k=1}^{12} 3 \left(\frac{1}{4} \right)^{k-1}$$

- 10 pts 13. Find the sum of the infinite geometric series

$$\sum_{k=1}^{\infty} -\frac{9}{4} \left(-\frac{4}{9}\right)^{k-1}$$

- 5 (bonus) 14. Compute the derivative of $f(x) = x^2 - 4$ using using *only* the limit definition of the derivative,

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$